

■ 4.2.3 FORTRAN

Mathematica and FORTRAN differ in many ways. Nevertheless, some superficial features, such as the way you enter basic arithmetic operations, are quite similar.

Like any modern computer language, *Mathematica* does not care what column you enter your input in. Nor does it care how many letters are in your variable names.

FORTRAN	<i>Mathematica</i>
$f(x)$	$f[x]$
$x*y$	$x*y$ or $x\ y$
$x**y$	x^y
$\text{sqrt}(x)$, $\text{alog}(x)$ etc.	$\text{Sqrt}[x]$, $\text{Log}[x]$ etc.
$a(i)$ (array reference)	$a[[i]]$
$\text{do label } i = \text{imin}, \text{imax} \leftarrow \text{statement} \leftarrow \text{label continue}$	$\text{Do}[\text{statement}, \{i, \text{imin}, \text{imax}\}]$
$\text{if}(i.\text{eq}.j) \text{ statement}$	$\text{If}[i == j, \text{statement}]$

Some correspondences between FORTRAN and *Mathematica*.

You can get *Mathematica* output in FORTRAN input form using the *Mathematica* function `FortranForm`.